

Perpetual Futures Pricing with Stochastic Rates and Clamp*

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Abstract

We develop, characterize, estimate, and evaluate a no-arbitrage model of perpetual futures prices with a clamp function and stochastic interest rates. Driven by DeFi interest rates, the model reproduces the salient fluctuations in futures-to-spot ratios. At higher frequencies, these ratios exhibit a bimodal distribution; the model matches this feature and explains it through the inaction region induced by the clamp. The relation between clamping strength and funding-rate volatility is nonmonotonic. The case with constant interest rates is solved in closed form, solution multiplicity is shown to be possible, and a computational algorithm is presented that verifies uniqueness for the results.

Keywords: Cryptocurrencies, derivatives. JEL codes: E12, G13.

1 Introduction

Perpetual futures, introduced by digital-asset exchanges, have now traded for several years. Unlike standard fixed-maturity futures, perpetuals do not need to be rolled and concentrate liquidity in a single contract. A potential drawback is that, without a final settlement date, futures prices could drift away from spot. Exchanges address this with funding payments, often specified with a clamp that makes the funding rate less sensitive when futures and spot are very close and more sensitive when deviations are large. Experience to date suggests this mechanism mostly keeps futures prices near spot, though the extent and conditions under

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Figure 1: Log futures-to-spot ratios for perpetuals and finite maturity futures. All series are hourly close prices, with the BRTI index as the BTCUSD spot rate. Data for perpetuals is from Binance, data for the CME contract from LSEG.

which it does so remain to be assessed systematically. For an example of the behavior of perpetual prices, consider Figure 1 that shows hourly close log futures-to-spot ratios for perpetuals from Binance alongside futures-to-spot ratios for finite maturity contracts from Binance and the CME. Not only are perpetual futures prices typically very close to spot, but that relation is stationary in contrast to the well-known convergence paths for finite maturity contracts. This paper studies the behavior of perpetual futures prices both empirically and through a no-arbitrage model.

We begin by documenting properties of the futures-to-spot ratio for two of the most actively traded Binance contracts, BTCUSDT and ETHUSDT. Since 2020, the hourly close price of each perpetual deviated from its corresponding spot price by more than 1% fewer than ten times out of over 45,000 hourly observations, underscoring how closely perpetuals have tracked spot despite pronounced underlying volatility.

We present a no-arbitrage model with clamp-style funding rule that mirrors exchange practice. An efficient algorithm is developed for solving the model and verifying uniqueness

of computed solutions. The case with constant interest rates is solved in closed form. The quantitative model is cast as a one-factor specification in the interest-rate ratio (numeraire versus base currency) and is calibrated using daily on-chain interest rates, together with futures data. The model can reproduce the salient fluctuations of the empirical futures-to-spot ratios. At higher frequencies, the log futures-to-spot ratio exhibits a bimodal distribution; the model matches this feature and explains it as a consequence of the clamp’s inaction region.

The quantitative model is also used to evaluate the design of the contract. Surprisingly, reducing the strength of the clamping effect does not necessarily make funding rates more volatile. This happens because futures-to-spot ratios endogenously respond to the funding rate specification. Our experiments suggest that it might be worthwhile for exchanges to explore further refinements in the clamp function.

The paper’s main contribution is the development of a no-arbitrage model with a clamp and the study of the fluctuations in the futures-to-spot ratio. The paper extends the exclusively theoretical analysis in Akerer, Hugonnier, and Jermann (AHJ, 2025) that did not include clamps. Other papers have studied perpetuals prices from the perspective of no-arbitrage settings, namely, He et al. (2023), and Dai, Li, and Yang (2025) who derive no-arbitrage bounds with a clamp. Their models have constant interest rates and effectively constant futures-to-spot ratios. Angeris et al (2022) derive the funding rate value such that the perpetuals price is explicitly given by a function of the spot price and the unobservable parameters of the prices process. Kim and Park (2025) also derive funding rate specifications. Several studies have focused on empirical properties of perpetuals prices with less formal no-arbitrage models: Alexander et al (2020), Christin et al (2023), Schmeling, Schrimpf and Todorov (2022), and Gornall, Rinaldi and Xiao (2024). There seems to be no prior work that highlights the bimodal nature of the distribution of futures-to-spot ratios and its connection to the clamp function, nor the potential for multiplicity with a clamp.

2 Data

Denote the perpetual futures and spot prices by f_t and x_t , respectively. We start by documenting the behavior of the ratio f_t/x_t for hourly close prices covering 2020 to 2025.6 together with the associated funding payments. Funding payments at Binance are due every 8 hours so that the cash flow for the long side of the contract for a funding period is

$$f_{t+1} - f_t - F_{t+1},$$

with the funding payment approximately equal to

$$F_{t+1} \approx \underbrace{\left(\kappa \left(\frac{f_t - x_t}{x_t} \right) + \iota \right)}_{\text{funding rate}} x_{t+1}, \quad (1)$$

with coefficients $\kappa > 0$ and ι . Most important is the difference between the futures and the spot price which anchors the futures price to the spot price. For instance, if $(f_t - x_t) > 0$ the long side needs to make a funding payment. This makes the long position less attractive and contributes to driving f_t closer towards x_t . The exact mapping from $(f_t - x_t)$ to the funding payment is exchange-specific. For instance, Binance averages over the 8-hour funding period and uses a nonlinear transformation, a so-called clamp function. This function is specified so that if futures and spot prices are very close, the funding rate only depends on ι but not on $(f_t - x_t)$. More details on the exact specification of the clamp function are given below.

The first row of Figure 2 displays time series for f_t/x_t for BTC and ETH in terms of USDT at hourly frequency. A strong common component in these two contracts is evident. It is worth noting how close futures prices have been to spot prices over this period, considering also that this market has been open 24-7 and that the underlying spot prices have been extremely volatile.

As shown in the figure, the future-to-spot ratios seem to spend considerable time at a relatively constant level slightly below 1 interspersed with extended bursts of higher values. The second row displays the funding rates determined at 8-hour intervals. As implied by their specification, funding rates track future-to-spot ratios. Binance has set the interest component, ι , to a constant 0.0001 per 8-hour period, corresponding approximately to 10% in annual terms ($3 \times 365 \times 0.0001 \simeq 0.11$). As is visible in the figure, there is a region of inaction around this value, which is induced by a nonlinear clamp function.

Figure 3 presents histograms for $\ln(f_t/x_t)$ and funding rates. First, the log futures-to-spot ratios have a bimodal hourly frequency distribution. The concentrated frequency mass on the left – around -0.0005 – was visible in the time series, the one on the right becomes visible in this plot. Second, funding rates display a very high point mass at 0.0001 which is induced mechanically by the clamp function.

Table 1 displays select statistics. Consistent with the visual inspections, median $\ln(f_t/x_t)$ values are below the mean for both currencies, at close to -0.0003 . Averaging over eight hourly periods reduces standard deviations moderately.

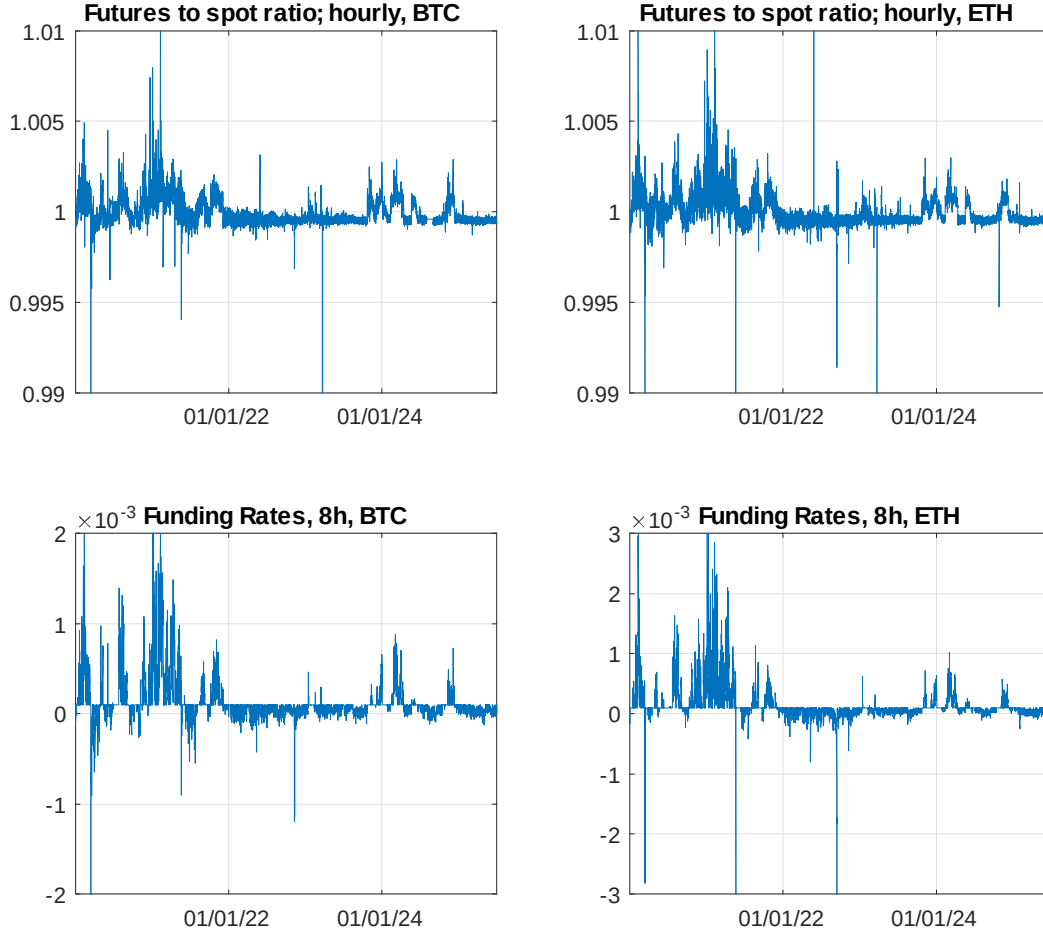


Figure 2: Futures-to-spot ratios and corresponding funding rates. Data covers the BTCUSDT and ETHUSDT contracts on Binance.

Statistic	BTC	ETH
mean $\ln(f/x)$	-0.000056	0.000015
med $\ln(f/x)$	-0.000319	-0.000294
std $\ln(f/x)$	0.000654	0.000735
Corr($\ln(f/x), \ln(f/x)(-1)$)	0.880882	0.862869
std ma($\ln(f/x), 8$)	0.000606	0.000675
approximate κ	0.25	0.31
freq(abs($\ln(f/x)$) > 0.001)	0.072	0.099
freq(abs($\ln(f/x)$) > 0.01)	0.00015	0.00013

Table 1: Statistics based on hourly close prices of BTCUSDT and ETHUSDT perpetuals on Binance for 2020 to 2025.6.

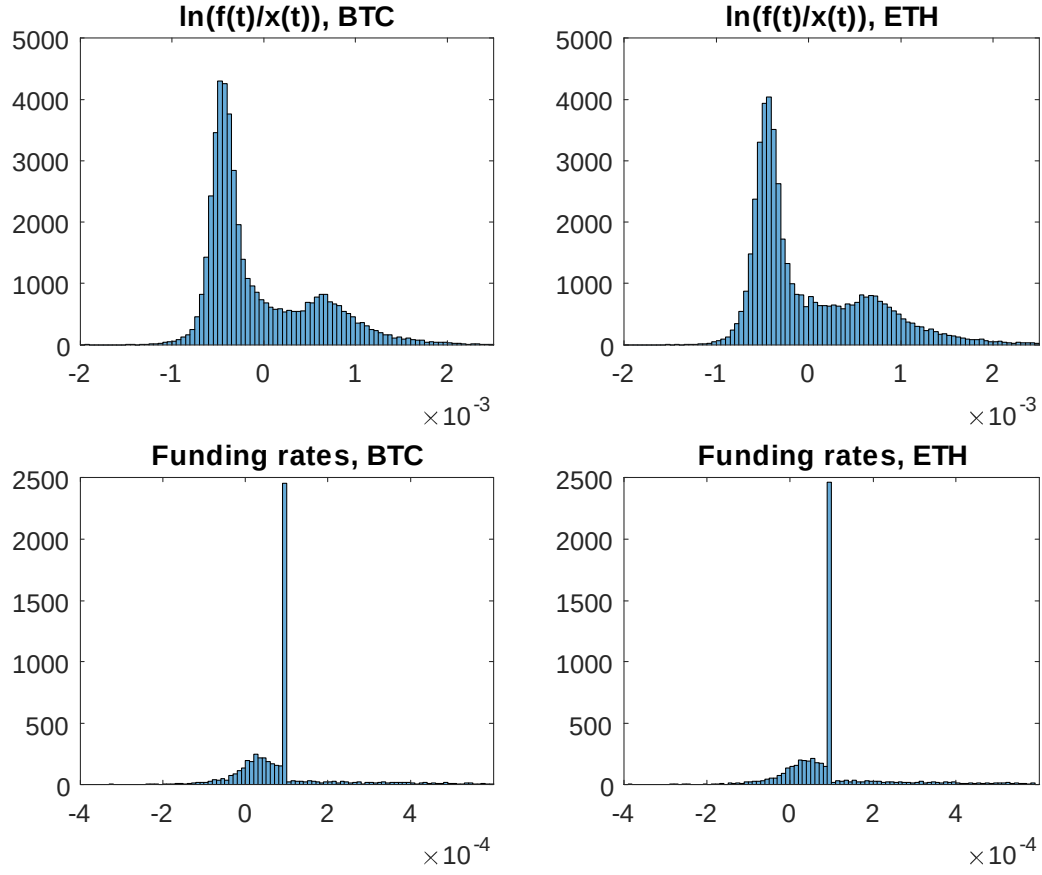


Figure 3: Histograms for futures-to-spot ratios (hourly) and funding rates (at 8-hour intervals). Data is for the BTCUSDT and ETHUSDT contracts on Binance.

3 No-arbitrage pricing

This section presents a no-arbitrage pricing model—building on the basic specification in AHJ (2025)—cast in a recursive setup that conveniently accommodates stochastic interest rates. The model is then extended to include a clamp function mimicking Binance’s specification. We present closed form solutions for special cases for each model, and we provide some initial quantitative perspectives.

3.1 Basic model

Time is discrete. Uncertainty is represented by a finite-state Markov chain with the realization of $z \in \{z_1, \dots, z_N\}$ at each time period. Following standard no-arbitrage principles, assume a stochastic discount factor $m(z, z')$, with z' the next-period realization of the state. Ruling out arbitrage implies

$$E_{z'|z} m(z, z') \left[f' - f - \left(\kappa \left(\frac{f - x}{x} \right) + \iota \right) x' \right] = 0, \quad (2)$$

because entering the futures contract has no up-front cost in the current period. Rearranging and defining $\Gamma(z, z') \equiv x'/x$ and $f_x \equiv f/x$,

$$f_x = \frac{\frac{1}{E_{z'|z} m(z, z')} E_{z'|z} m(z, z') \Gamma(z, z') [\kappa - \iota]}{1 + \frac{1}{E_{z'|z} m(z, z')} E_{z'|z} m(z, z') \Gamma(z, z') \kappa} + \frac{\frac{1}{E_{z'|z} m(z, z')} E_{z'|z} m(z, z') [\Gamma(z, z') f'_x]}{1 + \frac{1}{E_{z'|z} m(z, z')} E_{z'|z} m(z, z') \Gamma(z, z') \kappa}.$$

Assuming the existence of arbitrage-free one-period risk-free claims

$$\frac{1}{E_{z'|z} m(z, z')} E_{z'|z} [m(z, z') \Gamma(z, z')] = \frac{1 + r_{\$}(z)}{1 + r_c(z)} \equiv I(z) \quad (3)$$

where $r_{\$}(z)$ and $r_c(z)$ the one-period interest rates in the numeraire (dollar) and the base currency. Indeed, this is the ratio of a one-period futures/forward price to the spot rate as implied by Covered Interest Parity. Specifically, the no-arbitrage equation for a one-period contract is given by $E_{z'|z} m(z, z') [x' - f^{(1)}] = 0$, which implies $(f^{(1)}/x)(z) = \frac{1+r_{\$}(z)}{1+r_c(z)}$.

Define the following recursive operator $T(f_x)(z)$ for the futures-to-spot ratio

$$T(f_x)(z) = \frac{I(z) [\kappa - \iota]}{1 + I(z) \kappa} + \left(\frac{1}{1 + I(z) \kappa} \right) \frac{1}{E_{z'|z} m(z, z')} E_{z'|z} [m(z, z') \Gamma(z, z') f_x(z')]. \quad (4)$$

A fixed point, $f_x^* = T(f_x^*)$, satisfies the futures price in Equation (2) and, under some regularity conditions, a solution to the pricing equation is a fixed point of this operator. Given

processes for $m(z, z')$ and $\Gamma(z, z')$ one can derive futures-to-spot ratios $f_x(z)$ by iterating on Equation (4). If κ is large enough, the operator satisfies the discounting property and therefore has a unique fixed point.

For the special case of constant interest rates, $\frac{1+r_{\S}(z)}{1+r_c(z)} = \frac{1+r_{\S}}{1+r_c} = I$, Equation (4) becomes

$$T(f_x)(z) = \frac{I[\kappa - \iota]}{1 + I\kappa} + \left(\frac{1}{1 + I\kappa} \right) \frac{1}{E_{z'|z}m(z, z')} E_{z'|z} [m(z, z') \Gamma(z, z') f_x(z')].$$

Starting the iteration from $f_x(z') = f_x$, $T(f_x)(z) = T(f_x)$, and with κ high enough ($\kappa > 1 - 1/I$), the unique fixed point is

$$f_x = \frac{I[\kappa - \iota]}{1 + I\kappa} / \left[1 - \frac{I}{1 + I\kappa} \right] = \frac{\kappa - \iota}{\kappa - \frac{r_{\S} - r_c}{1 + r_{\S}}}. \quad (5)$$

This is consistent with the pricing formula derived in AHJ (2025). It differs slightly from the base case in AHJ because the funding here is indexed by x' as opposed to x . The specification used here is more in line with the contract used on Binance.

3.2 Clamp model

Many exchanges use variants of a clamp function. We now modify the model to include a function that mimics the specification used on Binance. Specifically, the cash flow to the long side is given by

$$f_{t+1} - f_t - r_t^f x_{t+1}$$

with the funding rate, r_t^f , defined as

$$r_t^f = r_t^p + \text{clamp}(\iota_t - r_t^p, -\eta, \eta)$$

and the premium index

$$r_t^p = \frac{f_t - x_t}{x_t}.$$

With the clamp, the funding rate is given as

$$\begin{aligned} r_t^f &= r_t^p + \eta && \text{for } (\iota_t - r_t^p) > \eta \text{ ("futures are low")} \\ r_t^f &= \iota_t && \text{for } |\iota_t - r_t^p| \leq \eta \text{ ("futures and spot are close")} \\ r_t^f &= r_t^p - \eta && \text{for } (\iota_t - r_t^p) < -\eta \text{ ("futures are high")} \end{aligned} \quad (6)$$

Typically, $\eta = 0.0005$ and $\iota = 0.0001$. Intuitively, the clamp induces a region of inaction

when the futures and spot prices are close, and for larger deviations, the transmission to funding rates is reduced. Binance computes the premium index over the 8-hour period as the average of r_t^p observed at 5-second intervals. In our model, a period is 8 hours, and the premium index is computed based on the r_t^p observed in the period before.

The futures-to-spot ratio implied by the absence of arbitrage satisfies the following conditions:

$$\begin{aligned} f_x &= \frac{I(z)}{1+I(z)} [1-\eta] + \frac{1}{1+I(z)} E_{z'|z}^Q [\Gamma(z, z') f'_x] & \text{for } (\iota(z) - (f_x - 1)) > \eta \\ f_x &= -I(z) \iota(z) + E_{z'|z}^Q [\Gamma(z, z') f'_x] & \text{for } -\eta \leq (\iota(z) - (f_x - 1)) \leq \eta \\ f_x &= \frac{I(z)}{1+I(z)} [1+\eta] + \frac{1}{1+I(z)} E_{z'|z}^Q [\Gamma(z, z') f'_x] & \text{for } (\iota(z) - (f_x - 1)) < -\eta \end{aligned} \quad (7)$$

with $E_{z'|z}^Q [\Gamma(z, z') f'_x] \equiv \frac{1}{E_{z'|z} m(z, z')} E_{z'|z} [m(z, z') \Gamma(z, z') f'_x]$.

3.2.1 Model solution

To solve the model, we define a recursive operator whose solution satisfies the no-arbitrage restrictions on the futures-to-spot ratio in Equation (7). This operator is not generally a contraction. Intuitively, the middle region of the clamp is problematic: when $I > 1$, there is no discounting. Using monotonicity, we show how to verify uniqueness of a computed solution and we characterize a special case in closed form.

The model is defined by an operator T on a function, or vector, $f_x(z)$:

$$T(f_x)(z) = T^{(j)}(f_x)(z), \quad (8)$$

where $j = j(z; f_x)$ is determined by the selector variable

$$G(z) \equiv 1 + [1 + I(z)] \iota(z) - E_{z'|z}^Q [\Gamma(z, z') f_x(z')].$$

The three regions are

$$G(z) > u \quad \Rightarrow \quad j = 1 \quad (\text{"low futures"}),$$

$$l \leq G(z) \leq u \quad \Rightarrow \quad j = 2 \quad (\text{"middle"}),$$

and

$$G(z) < l \quad \Rightarrow \quad j = 3 \quad (\text{"high futures"}).$$

The operators associated with these regions are

$$\begin{aligned}
T^{(1)}(f_x)(z) &= \frac{I(z)}{1+I(z)}[1-u] + \frac{1}{1+I(z)}E_{z'|z}^Q[\Gamma(z, z')f_x(z')] \\
T^{(2)}(f_x)(z) &= -I(z)\iota(z) + E_{z'|z}^Q[\Gamma(z, z')f_x(z')] \\
T^{(3)}(f_x)(z) &= \frac{I(z)}{1+I(z)}[1-l] + \frac{1}{1+I(z)}E_{z'|z}^Q[\Gamma(z, z')f_x(z')].
\end{aligned} \tag{9}$$

The symmetric case for the clamp used elsewhere in the paper has $\eta = u = -l$. The model is solved by iterating from some initial condition for $f_x(z')$ until convergence to a fixed point $f_x(z) = T(f_x)(z)$. Convergence is typically fast because regions 1 and 3 imply strong discounting.

No-arbitrage requires

$$E_{z'|z}^Q \Gamma(z, z') = I(z) = \frac{1+r^s(z)}{1+r^c(z)}.$$

The following properties allow efficient verification of uniqueness. We use monotonicity of the recursion and min-max initial conditions, that is, we start the iteration from a very high and a very low initial function $f_x(z')$. If both recursions converge to the same fixed point, this fixed point is unique.

Lemma 1 (*Monotonicity*): *The operator T is order-preserving: for any $f, g : Z \rightarrow \mathbb{R}$,*

$$f \leq g \implies T(f) \leq T(g)$$

(*componentwise*).

Proof. Fix z , if $f \leq g$ the conditional expectation common in all three regions of the recursion is larger for g because state-price probabilities are nonnegative. If the region for z does not change $T(f) \leq T(g)$ for this z . Given the definition of $G(z)$, region numbering can only increase. Moving from 1 to 3 increases the function value because $-l > -u$; a little algebra shows that transitions from 1 to 2 and from 2 to 3 are continuous. Therefore, without down-jumps for region transitions we have $T(f) \leq T(g)$. ■

Proposition 2 (*Uniqueness on an interval*): *Let $D = [\underline{f}, \bar{f}]$ be a bounded interval of vectors, ordered componentwise, and suppose that T maps D into itself and is monotone and continuous. Start iterating T from $\underline{f}$ and $\bar{f}$. If both sequences converge to the same limit, then that limit is the unique fixed point of T in D .*

Proof. Let $\underline{f}_0 = \underline{f}$ and $\bar{f}_0 = \bar{f}$. Since $T(D) \subseteq D$ and T is monotone, the lower sequence $\underline{f}_{n+1} = T(\underline{f}_n)$ is increasing, the upper sequence $\bar{f}_{n+1} = T(\bar{f}_n)$ is decreasing, and both remain in D . Hence both sequences converge. By continuity of T , their limits are fixed points. Any fixed point $f^* \in D$ satisfies $\underline{f}_n \leq f^* \leq \bar{f}_n$ for all n . Therefore, if the two limits coincide, every fixed point in D must equal the common limit. ■

Uniqueness is only guaranteed in the designed interval. We use specific properties of the problem to impose lower and upper bounds. For the calibration used in the paper, starting from low and high initial conditions both converge to the same fixed point.

Constant interest rates and funding parameters To build intuition about the solution and potential multiplicity, consider the model under constant interest rates and constant funding parameter. The following proposition characterizes this case.

Proposition 3 (*Constant-rate case*) Assume $I(z) = I$ and $\iota(z) = \iota$. Suppose $I \neq 1$, and define the exogenous selector,

$$\chi \equiv 1 + \frac{\iota}{1 - I}.$$

Then futures-to-spot ratios are characterized as follows.

If $1 - I > 0$,

$$f_x = \begin{cases} I(1 - \eta), & \chi > \eta \\ -\frac{I}{1 - I}\iota, & -\eta \leq \chi \leq \eta \\ I(1 + \eta), & \chi < -\eta \end{cases}$$

In this case, χ selects a unique solution.

If $1 - I < 0$,

$$f_x = \begin{cases} I(1 - \eta), & \chi < \eta \\ -\frac{I}{1 - I}\iota, & -\eta \leq \chi \leq \eta \\ I(1 + \eta), & \chi > -\eta \end{cases}$$

and if

$$\iota \in [(1 - \eta)(I - 1), (1 + \eta)(I - 1)],$$

there are multiple solutions, with three solutions in the interior of the interval and two solutions at the endpoints; otherwise, the solution is unique.

If $1 - I = 0$, $f_x = 1 - \eta$ if $\iota > 0$, $f_x = 1 + \eta$ if $\iota < 0$, and for $\iota = 0$, f_x can take any value in the middle range.

Proof. Iterating equation (8) from a non-state-contingent vector $f_x(z) = f_x$ assuming $I(z) = I$ and $\iota(z) = \iota$ implies that $T(f_x)$ is non-state-contingent, and if there is a unique

fixed point it is also non-state-contingent. The within-region fixed points for f_x are obtained by direct computation. The values for f_x in the region selectors are then replaced by the solution and the inequalities rearranged to express them in terms of the common selector variable χ . For $I < 1$, the graph of $T(f_x)$ as function of f_x , has slope smaller than 1 in each region, and is upward sloping (monotonic). At the two cutoff levels for the selector, $T(f_x)$ is continuous. Therefore, there is at most one solution; the solution is the fixed point selected among the three within-region fixed points. When $I > 1$, the graph of $T(f_x)$ as function of f_x , has a slope that is larger than 1 in the middle region and slopes that are smaller than one in the outer regions. Therefore, there can be three solutions. If the within-region fixed point of the middle region is not selected, one of the outer-region fixed points is the unique solution and is selected by χ . The range for ι that admits multiple solutions is directly implied by the range for χ corresponding to the middle region. ■

The pricing formula in the outer ranges when $f_x \in \{I(1 - \eta), I(1 + \eta)\}$ is closely related to the price of the basic (linear) model derived in Equation (5) above, $f_x = (\kappa - \iota) / \left(\kappa - \frac{r_s - r_c}{1 + r_s} \right)$. The clamp implies $\kappa = 1$ and in the outer regions $\iota = 0$; with these values, for the basic model, $f_x = \frac{1 + r_s}{1 + r_c} = I$. The parameter η "clamps" the funding rate toward 0 in the outer regions. So, for instance, for a price in the lower region $f_x = I(1 - \eta)$, the long position receives funding payments which the clamp reduces by η . Therefore, due to the clamp, the long position receive less in funding payments (in every period) and the futures price is reduced by the multiplicative factor $(1 - \eta)$.

For a numerical example illustrating the empirically relevant case, set $\iota = 0.0001$ per 8-hour period and $\eta = 0.0005$, consistent with the Binance parameterization. For our sample period, $I \approx 1.00004$ on average across the two contracts. This implies $\chi = 1 + \frac{\iota}{1 - I} \approx -1.5$, which selects the first region and $f_x = I(1 - \eta) = 1.00004 \times 0.9995 = 0.99954$, corresponding to $f_x - 1 \approx -4.6$ bps. This is close to the high mass point of the empirical futures-to-spot distribution in Figure 3. Since only one region is selected, the solution is unique.

As stated in the proposition, if $1 - I < 0$, uniqueness is not guaranteed. For illustration, set $\iota = I - 1$ with $I > 1$. Then $\chi = 0$, so all three regions are selected. For $I = 1.00004$ and $\eta = 0.0005$, the three solutions are

$$f_x \in \{0.99954, 1.00004, 1.00054\}.$$

The interval of values for ι that generates multiple solutions is small. In particular

$$\iota \in [(1 - \eta)(I - 1), (1 + \eta)(I - 1)] = [0.00003998, 0.00004002].$$

There are three solutions in the interior of this interval and two solutions at the endpoints.

4 Calibration

Based on Equation (3), a convenient empirical specification is a one-factor model in the interest ratio, such that

$$\frac{1}{E_{z'|z} m(z, z')} [m(z, z') \Gamma(z, z')] = \frac{1 + r_s}{1 + r_c}(z) \exp \left(g_z \left(\frac{1 + r_s}{1 + r_c}(z') \right) \right) \quad (10)$$

for some function $g_z(\cdot)$ with the restriction

$$E_{z'|z} \exp \left(g_z \left(\frac{1 + r_s}{1 + r_c}(z') \right) \right) = 1. \quad (11)$$

This specification conflates the typical SDF factor with the exchange rate factor; in a one-factor model this is without loss of generality. A more standard approach would be to explicitly specify separate processes for $m(z, z')$ and $\Gamma(z, z')$, but that would be less parsimonious. For a first pass, the more transparent approach seems preferable.

The interest ratio is specified as a Markov chain approximating an AR(1) process with normal innovations. This requires values for the mean μ_I , standard deviation σ_I , and AR(1) coefficient ρ_I . The function $g_z(\cdot)$ is assumed to be linear in the normalized innovation of the interest ratio with a constant factor loading b_z . There are also state-contingent constant $a_z(z)$ which adjust for the convexity of the exponential function to guarantee that Equation (11) holds. In total, the calibration, therefore, requires four parameters: $(\mu_I, \sigma_I, \rho_I, b_z)$.

The parameters are set to target four sets of moments. The mean and standard deviation of the interest factors, μ_I and σ_I , are set to match daily means and standard deviations of the empirical counterparts based the interest rates for USDT, BTC, and ETH from Aave. The persistence parameter ρ_I is set to match the equally-weighted average of the first-order and seventh-order serial correlation coefficients of the empirical counterparts. The empirical autocorrelations of the interest ratios decay more slowly than a standard AR(1) process, and this approach strikes a compromise in matching the relatively higher and lower frequency behavior. The factor loading b_z is set to match the median of the empirical futures-to-spot ratios. The median is preferred over the mean as being more empirically robust given the shape of the empirical distribution displayed in Figure 3.

Daily interest rates are computed as the realized percentage change in Aave's *liquidityIndex* over the 24-hour period. This measures the realized return a depositor receives over a 24-hour period. At the beginning of the sample in 2020, Ethereum V2 was the dominant

Parameter	Symbol	Value	
mean interest factor	μ_I	$3.7175e - 05$	
std interest factor	σ_I	$3.1271e - 05$	
persistence interest factor	ρ_I	0.972	
factor risk loading	b_z	0.52	
Target moment	Model	Data BTC	Data ETH
mean interest factor daily	0.000112	0.000127	$9.63e - 05$
std interest factor daily	$9.2e - 05$	$9.1e - 05$	$9.7e - 05$
ρ_1 interest factor daily	0.94	0.82	0.84
ρ_7 interest factor daily	0.55	0.64	0.68
median futures-to-spot -1 daily	-0.000342	-0.000361	-0.000322
Model fit		BTC	ETH
R2med		0.56	0.57
R1med		0.24	0.31

Table 2: Calibration and model fit. R2med equals 1 minus the sum of the squared deviations from the model divided by the sum of the squared deviations from the median. R1med uses absolute deviations.

protocol. In 2025, activity had mostly migrated to Ethereum V3. We switch the data source from V2 to V3 on January 8, 2024. On this day, the supply for all three assets considered, USDT, BTC and ETH, on V3 had overtaken the supply on V2.¹ Observations below 0.9998 and above 1.0006 are removed; this removes 12 outliers out of total of 1716 observations.

A model period is 8 hours to match the funding-payment frequency at Binance. Interest rates from Aave are available at daily frequency, and hourly close prices for futures and spot rates are averaged to daily frequencies.

Parameter values and calibration targets are displayed in Table 2. Given that we directly specify the process for the interest ratio, the parameters μ_I and σ_I are simply set to 1/3 of the target moment which is taken to be the average for BTC and ETH (three 8-hour model periods are aggregated up to the daily target). One could calibrate the model separately for BTC and ETH, but the differences would not be beyond standard errors. The persistence parameter ρ_I is set to minimize the sum of the squared deviations of the implied daily serial correlation coefficients from one to seven lags. Targeting a range as opposed to just a given horizon is chosen because the empirical serial correlations over this range decay less than for a standard AR(1). For a given interest process, the factor loading, b_z , is set for the median futures-to-spot ratios of the model to match its empirical counterparts. A larger positive value for b_z increases the future-to-spot ratio as this increases the positive covariance between the innovation in the pricing factor and the f_x ratio in the pricing equation, Eq (4).

¹V3 supply first exceeded V2 supply on October 18, 2023 for USDT, January 8, 2024 for WBTC, and October 8, 2023 for WETH.

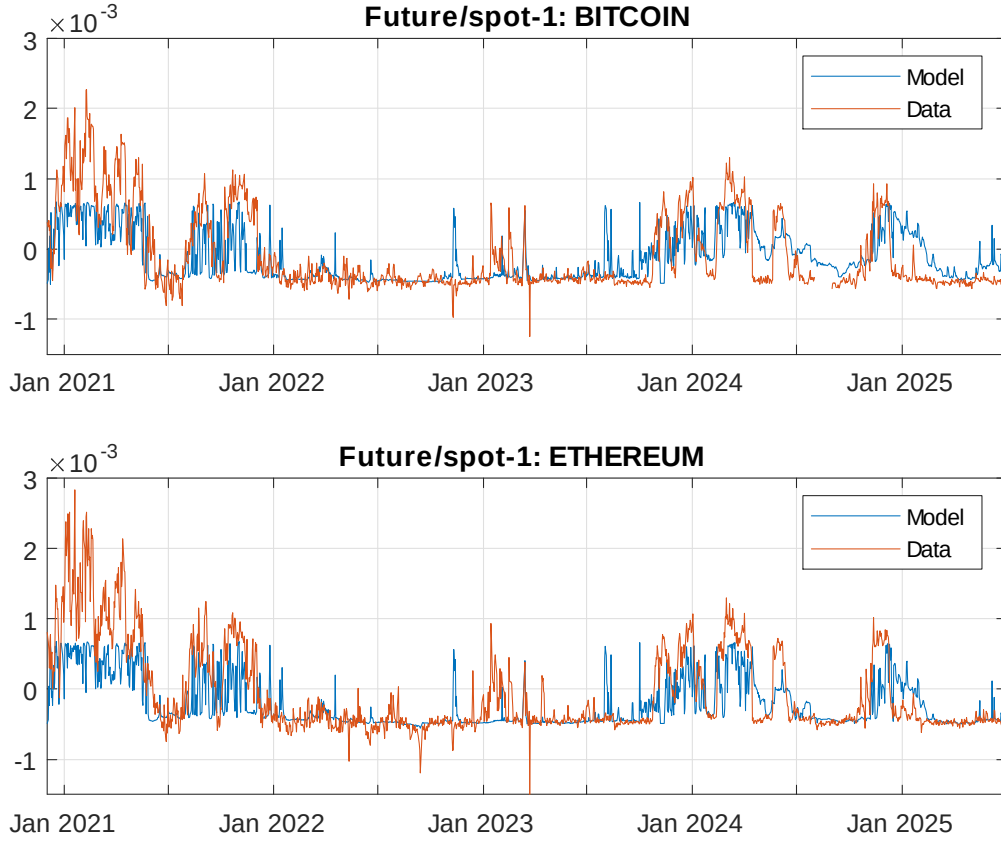


Figure 4: Futures-to-spot ratios in the model and the data. Model-implied futures ratios are generated with interest rates from Aave.

5 Model implications

To evaluate the model’s ability to explain fluctuations in futures ratios, we simulate the model with the empirical interest rates for USDT/BTC and USDT/ETH. The pricing function $f_x(z)$ is linearly interpolated with respect to the empirical counterpart of the interest ratio $I(z)$. Specifically, the interest ratio for a given day (which is available daily), I_t , is converted to an 8-hour period as $(I_t - 1)/3 + 1$ and the model-implied f_t/x_t is compared to the daily average f_t/x_t . The realized simulation paths, for daily observations, are displayed in Figure 4 alongside their empirical counterparts. The several extended periods with $f_t/x_t - 1$ above zero are approximately replicated by the model; the model also replicates the extended periods when futures ratio stabilized slightly below zero. The model cannot produce the very high futures ratios observed during 2021. This is consistent with a view that arbitrage linkages across markets were less developed in the early days of DeFi lending.

As a formal measure of goodness of fit, consider an R-squared metric defined as

$$1 - \sum (f_{x,t}^{data} - f_{x,t}^{model})^2 / \sum (f_{x,t}^{data} - \text{median} f_{x,t}^{data})^2.$$

According to this measure, the model explains 56% of the deviations from their medians for BTC and ETH. Measuring goodness of fit with respect to the empirical median as opposed to the mean is motivated by the fact that the calibration targets the median of the futures-to-spot ratios. Therefore, a value of 0 would imply that the model has no explanatory power beyond the median. As alternatives, Table 2 also reports absolute deviations compared to the median.

Figure 5, in the top two panels, displays histograms for daily values of $(f_t/x_t - 1)$ from the data and the model. As shown earlier, empirical futures ratios have a pronounced bimodal distribution. The model-implied futures ratios approximately replicate this property. To appreciate the effect of the model, consider that third row which presents the distribution of the interest ratios used to simulate the model. The distribution produced by the model for $(f_t/x_t - 1)$ is a lot more stretched to the right than the distribution of the interest ratio. The red bars included in the upper panels delimit the range of inaction of the clamp. Even without considering the model, the placements of the red bars in the first row strongly suggest that the clamp matters for the bimodality.

The theoretical pricing function in Figure 6 provides an explanation for this property. The steep region of the pricing function corresponds to the range of inaction of the clamp. In the inaction region of the clamp, the stabilizing effect of the funding payments is turned off locally. As a consequence, probability mass of f_t/x_t in the inaction range is reduced relative to the outer ranges. The slope of the pricing function in the outer regions is not far from one, while it is about an order of magnitude larger in the middle region. If the distribution of the interest factor is evenly spread across the three regions, a bimodal distribution for f_t/x_t results.

6 Design choices

This section explores the impact of variations in the specifications of the interest factor ι and the strength of the clamp action η . While the clamp keeps funding rates constant within the inaction region, we find that removing the clamp would lead to lower overall volatility in funding rates.

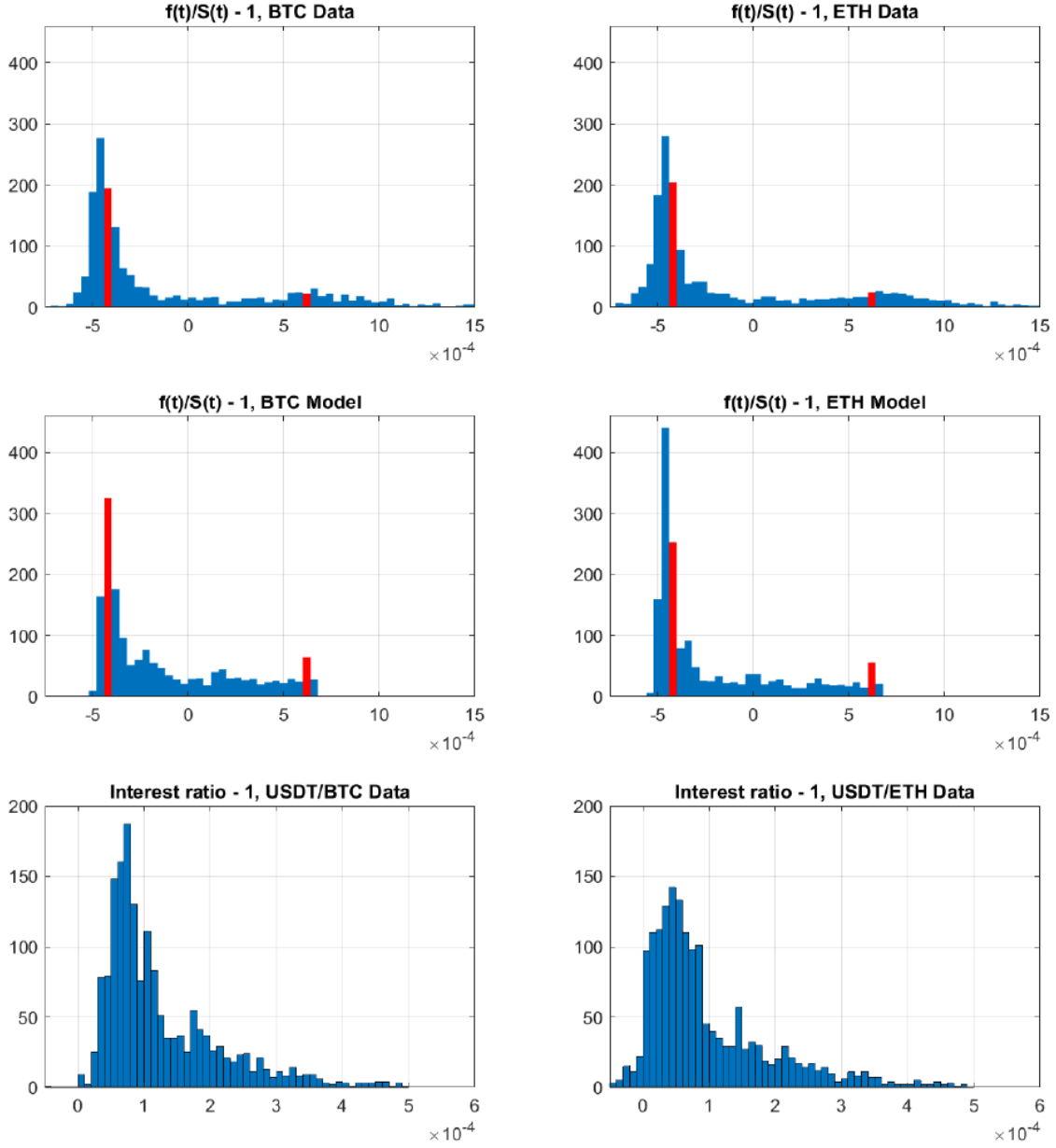


Figure 5: Futures-to-spot ratios from the data and the simulated model are in the upper two panels. The red bars indicate the limits of the clamping regions. The third panel displays the distribution of the interest ratios used to simulate the model. Displayed Interest ratios are for daily rates; the model is simulated with the corresponding 8-hour ratio derived as $1/3$ of the daily rates.

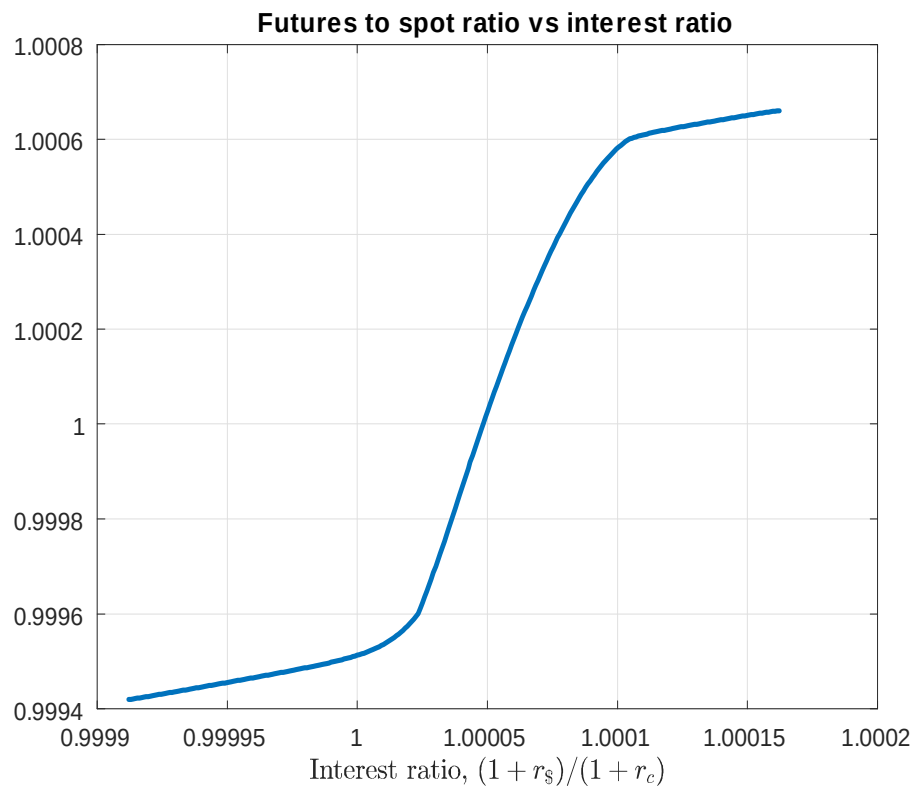


Figure 6: Model-implied pricing function

6.1 Clamping strength, η

Table 3 documents the impact of variations in the clamping parameter η . The clamp function, Equation 6, has two related effects when compared to the basic linear specification, Equation 1. First, the clamp introduces a region of inaction so that small deviations of the futures from the spot are not transmitted to the funding rate. Second, larger deviations between futures and spot are transmitted in a reduced way. The parameter η determines the extent of the inaction region as well as the extent of the reduction in the transmission of larger deviations of futures from spot to funding rates. Effectively, the clamp eliminates noise in funding rates and reduces extreme fluctuations conditional on given fluctuations in futures-to-spot ratios. Such stability can be attractive to investors, in particular, as this can limit liquidations induced by large funding payments. However, given that futures-to-spot ratios depend themselves on the design of the clamp function, it isn't a priori clear whether a larger (more aggressive) clamp parameter η ends up stabilizing funding rates or not.

The results in Table 3 show that the relation between η and the standard deviation of the funding rate is not monotonic. Specifically, the benchmark specification used by Binance is close to a local maximum for the volatility of funding rates. Changes in η from the benchmark in both directions can lower the volatility of the funding rate. Somewhat surprisingly, the case of $\eta = 0$ which eliminates the clamp not only lowers the volatility of futures-to-spot ratios but also of funding rates.² Increasing the impact of the clamp relative to the benchmark to $\eta = 0.001$ stabilizes funding rates and makes futures-to-spot ratios more volatile than the benchmark. Overall, this analysis suggests that current contracts could be further improved through alternative specifications of the clamp function or its elimination.

6.2 Interest factor, ι

In line with Corollary 1 in AHJ (2023), for the basic perpetual contract (with no clamp), setting $\iota(z) = (r_s(z) - r_c(z)) / (1 + r_s(z)) = 1 - 1/I(z)$ implies that $f_x(z) = 1$. For the clamp model, this specification produces three solutions. It is not a priori clear how such a finding should be interpreted. In any case, the parameter region that features this property is small (in line with our discussion in 3.2.1).

While theoretically straightforward, the practical implementation of this "ideal" perpetual could be challenging, as this would require real-time interest rates for each funding period. A simpler alternative that might a priori be promising could be to set the interest factor based on the average interest rate ratio, that is $\iota(z) = 1 + 1/E(I(z))$. Table 3 shows

²Setting $\eta = 0$ in the benchmark model with a clamp is equivalent to the basic linear model with $\kappa = 1$ and $\iota = 0$.

Design	$\text{mean}(f_x - 1)$	$\text{median}(f_x - 1)$	$\text{std}(f_x)$	$\text{std}(r^{\text{fund}})$
Benchmark				
Clamp $\iota(z) = 0.0001, \eta = 0.0005$	-0.00012	-0.00018	0.00034	3.9e-05
Basic	-0.00017	-0.00017	0.00010	
Change clamping, η				
Clamp $\eta = 0$	4.1e-05	4.1e-05	3.0e-05	3.0e-05
Clamp $\eta = 0.001$	0.00025	0.00032	0.00051	1.4e-05
Change interest factor, $\iota(z)$				
Basic, $\iota(z) = 1 - 1/I(z)$	0	0	0	
Clamp, $\iota(z) = 1 - 1/E(I)$,	0.00048	0.00054	0.00014	1.9e-05
Basic	3.9e-05	4.0e-05	0.00009	
Clamp, $\iota(z) = 0.00009103$	1.0e-07	1.1e-05	0.00035	3.1e-05
Basic	-0.00014	-0.00014	0.00010	

Table 3: Futures-to-spot ratios and funding rates depending on design choices for clamping strength η and interest factor $\iota(z)$.

that for the basic model this can make the mean futures-to-spot ratio, f_x , close to 1; however, the model with the clamp does not produce this outcome. Mean and median f_x are monotonic in the interest factor. Therefore, a target mean level for f_x of 1 can be achieved. For the calibrated model that happens approximately for $\iota(z) = 0.00009135$, see Table 3. Practically, of course, this value would have to be estimated and the target would therefore not be hit exactly.

7 Conclusion

Perpetual futures prices for BTCUSDT and ETHUSDT on Binance have tracked corresponding spot prices very closely between 2021 and 2025. In this paper, we present a no-arbitrage model with interest rate risk and a clamp function. An efficient computational algorithm that can deal with potential multiplicity is presented and the case with constant interest rates is solved in closed form. Driven by daily interest rates from Aave, the model reproduces salient properties of the time series of empirical futures-to-spot ratios. Empirical futures-to-spot ratios at high frequency display a bimodal distribution. Our model approximately matches this property and explains it as a consequence of the inaction region in the clamp function. Variations in the specification of the clamp function in the quantitative model demonstrate a non-monotonic relation between clamp strength and the volatility of funding rates. This suggests that it might be worthwhile for exchanges to revisit the specification of the clamp function.

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